Problem (math_baseline): 4 points

(b) Run your evaluation script on Qwen 2.5 Math 1.5B. How many model generations fall into each of the following categories: (1) correct with both format and answer reward 1, (2) format reward 1 and answer reward 0, (3) format reward 0 and answer reward 0? Observing at least 10 cases where format reward is 0, do you think the issue is with the base model's output, or the parser? Why? What about in (at least 10) cases where format reward is 1 but answer reward is 0?
Answer:
Temperature = 1.0, p = 1.0, 1 shot
Total format reward: 823.0
Total answer reward: 128.0
Total reward: 128.0
2.5% pass rate

Problem (sft_experiment): Run SFT on the MATH dataset

The SFT dataset mentioned in the doc "/data/a5-alignment/MATH/sft.jsonl" is not publicly available, so I'm using the reason traces from [“open-r1/OpenThoughts-114k-math”] (open source attempt to replicate deepseek r1). I only selected responses that are shorter than 1024 and answers are correct. Overly verbose reasoning trace is detrimental because the model may not even finish its thought before hitting the context length limit

Fine tuned on 86 examples, training loss less than 2
Temperature = 1.0, p = 1.0, 1 shot
Total format reward: 1415.0
Total answer reward: 639.0
Total reward: 639.0
12.7% pass rate

Fine tuned on all 784 examples, the training loss is ~2.
Total format reward: 1946.0
Total answer reward: 1003.0
Total reward: 1003.0
20% pass rate

Problem (expert_iteration_experiment): Run expert iteration on the MATH dataset (2points) (6 H100 hrs)

Run expert iteration on the MATH dataset (provided at /data/a5-alignment/MATH/train.jsonl)

using the Qwen 2.5 Math 1.5B Base model, varying the number of rollouts G per question and the number of epochs used in the SFT step, and using n_ei_steps = 5. Log the entropy of the model's responses over training. Make sure to have vLLM terminate generations at the second answer tag </answer>, as done in the SFT section.
Deliverable: Validation accuracy curves associated with different rollout configurations. Try at least 2 different rollout counts and epoch counts.
Deliverable: A model that achieves validation accuracy of at least 15% on MATH.
Deliverable: A brief 2 sentence discussion comparing to your SFT performance, as well as perfor-mance across EI steps.
Deliverable: A plot of the entropy of the model's responses over training.

Answer:
G=2, pick only one correct answer for each question (if there is a correct answer)
Iteration = 0, epoch=10, batch_size = 8, loss = 0.7, valid_accuracy = 46.4%
Iteration = 1, loss = 0.39, valida_accuracy = 55.2%
Iteration = 2, loss = 1.15, validation_accuracy = 57.2%
Iteration = 3, loss = 5.2, validation_accuracy = 57.4%

All validation set:
Total reward so far: 2737.0
Total format reward: 3887.0
Total answer reward: 2737.0
Total reward: 2737.0
Pass rate 54.7%

Problem (grpo_train_loop): GRPO train loop (5 points)

Reinforce with baseline:
Total format reward: 3791.0
Total answer reward: 1297.0
Total reward: 1297.0
Pass rate: 25.9%

Compare it with the TRL library from Huggingface (see hf_rl.ipynb). With the same hyperparameter settings and use_vllm = True, vllm_mode = "colocate":
Total reward so far: 2607.0
Total format reward: 4554.0
Total answer reward: 2607.0
Total reward: 2607.0
Pass rate: 52.1%

Problem (mmlu_baseline): 4 points

Zero shot Llama 3.1 8B on MMLU:
Total correct reward: 3819
Total incorrect reward: 5734
Total total_failed_to_extract: 4367
27% pass rate

Problem (mmlu_sft): 4 points

SFT on Llama 3.1 8B with safety_augmented_ultrachat_200k_single_turn data. 280 steps, loss 5344.0

Zero shot:
Total correct reward: 5923
Total incorrect reward: 5871
Total total_failed_to_extract: 2126
Pass rate: 42.5%

Problem (dpo_training): 4 points

I was able to run eval on SimpleSafetyTests and generated the output jsonl as required, but I wasn't able to use LLM as a judge to eval the answers from my model are harmful or not (it always judge the content as risky even when the generated text flat out refusing to answer the question). Therefore I had to endure “emotionally taxing”, read and judge the 100 answers myself.

Vanilla Llama 3.1 8B:
Some of the responses make no sense or gibberish, which makes it less helpful (in a good way I guess). But I'd say there is little to no safety built in at all. I'd consider ~90% of the answers as harmful

SFT model:
Now the model can call out if it is a crime or harmful activity and refuse to answer. I'd consider ~30% of the answers as harmful

DPO model (I'm using HuggingFace DPOTrainer):
Now the model refuses to answer the majority of the questions, so only maybe 5 - 10% answers are harmful. But the model also repeated a sentence again and again and kind of speaking in gibberish.

DPO model on MMLU eval:
Total correct reward: 447
Total incorrect reward: 1487
Total total_failed_to_extract: 11986
Pass rate: 3.2%